

SHARATH S RAO

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SUMMARY

GenAI Systems Engineer with 3+ years building production-grade LLM systems serving 24,000+ users. Specializes in multi-agent architecture (LangGraph), RAG pipelines, context engineering, and Evaluation-Driven Development. Proven track record shipping reliable AI systems using CI-integrated eval pipelines, LLM-as-Judge, and structured quality gates. Equally comfortable across AI systems, backend engineering, and QA in early-stage product environments.

TECHNICAL SKILLS

LLM Orchestration: LangChain, LangGraph, LangSmith, Ollama

Evaluation & Reliability: EDD, LLM-as-Judge, RAGAS, Promptfoo, DeepEval, CI-integrated Evals, Hallucination Detection, Guardrails

RAG & Retrieval: Retrieval-Augmented Generation, Semantic Search, Hybrid Search, MMR, HNSW, Embeddings, Chunking

Multi-Agent Systems: Multi-Agent Architecture, Agentic Workflows, Graph-based Workflow Orchestration

Vector Databases: PostgreSQL pgvector, Pinecone | **Open-Source Models:** DeepSeek, Llama, Ollama

Prompt & Context Engineering: Prompt Engineering, Context Engineering, Structured Outputs, Function Calling, Tool Calling

Full Stack & Infra: Python, NestJS, Next.js, REST APIs, PostgreSQL, Docker, System Design (HLD/LLD), Multi-tenancy

EXPERIENCE

Technical Head — AI, Full Stack & QA

Jan 2025 – Present

TestMySkills.ai · Zyni Innovations

Bangalore, India

- Multi-disciplinary founding engineer on a B2C/B2B skill assessment platform with multi-tenant architecture — scaled to 24,000+ users leading an 8-engineer team.
- Pioneered Evaluation-Driven Development (EDD):** CI-integrated eval pipeline with DB-stored failure cases replayed as pre-commit checks — enabling confident feature shipping 95%+ of the time.
- Built RAG pipelines achieving ~90% accuracy** via advanced chunking, embedding tuning, and MMR retrieval; semantic search optimized with PostgreSQL pgvector and HNSW indexing.
- Full Stack:** Next.js frontend, NestJS backend, multi-tenant architecture. **QA:** automated testing pipelines and coverage strategies across the platform.

AI Engineer

Jun 2024 – Jan 2025

Zysk Technologies

Bangalore, India

- Built the AI-based Skill Mastery Evaluation System** — automated LLM scoring and feedback serving 24,000+ users; **reduced latency 80%** (5 min → 1 min) via batching and parallel LLM execution.
- Implemented LLM routing by availability and cost; built LLM-as-Judge pipelines for hallucination detection and output verification; engineered fault-tolerant inference with retry, fallback, and guardrails.
- PushsendAI:** Designed personalized recommendation engine using reinforcement learning and matrix factorization; pilot commended by Veronica Beard (US fashion brand).

PROJECTS

Zyskie — AI-Powered HRMS Platform

2026

Zyni Innovations · Product Owner & Lead Engineer · Team of 8

- Built a GenAI-powered HRMS with LLM-driven hiring workflows, workforce management, and internal talent discovery.
- Designed full system architecture (HLD + LLD + UML); implemented **pgvector with HNSW and MMR** for semantic search; deployed **DeepSeek via Ollama** for local inference; integrated Zoho, Keka, BambooHR via n8n.

EDUCATION

B.Tech in Information Science & Engineering

2020 – 2024

JSS Academy of Technical Education

Bangalore, India

- CGPA: 8.8 / 10.0

ADDITIONAL

Publications: Hardware Inventory Management System Using IoT — IEEE (9 citations); Interactive Framework for Querying Data from Large PDFs — IEEE (7 citations); Traffic Control Management System for Emergency Vehicles — Telematique Journal
Open Source: Active contributor to the LangChain open-source ecosystem (langchain-ai).

Speaking: Keynote Speaker — IEEE Student Chapter Inauguration (150+ attendees), GenAI & metacognition; Prompt Engineering sessions at KSIT, JSS, SJCE Mysore, Presidency University (5+ sessions, 30–50 attendees each).

Awards: Emerging Leader Award (2025); Captain of Change Agents — Zysk Technologies.